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When Private Information Settles the Bill: Money and Privacy in Google's Market for Smartphone Applications

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Abstract. We shed light on a money-for-privacy trade-off in the market for smartphone applications (“apps”). Developers offer their apps at lower prices in return for greater access to personal information, and consumers choose between low prices and more privacy. We provide evidence for this pattern using data from 300,000 apps obtained from the Google Play Store (formerly Android Market) in 2012 and 2014. Our findings show that the market’s supply and demand sides both consider an app’s ability to collect private information, measured by the apps’s use of privacy-sensitive permissions: (1) cheaper apps use more privacy-sensitive permissions; (2) given price and functionality, demand is lower for apps with sensitive permissions; and (3) the strength of this relationship depends on contextual factors, such as the targeted user group, the app’s previous success, and its category. Our results are robust and consistent across several robustness checks, including the use of panel data, a difference-in-differences analysis, “twin” pairs of apps, and various measures of privacy-sensitivity and app demand.

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Keywords: Android • mobile applications • privacy • permissions • supply and demand for private information

1. Introduction

Striking a balance between “too little” and “too much” privacy protection in digital technologies and markets poses a nontrivial challenge for regulators and society as a whole. On the one hand, the market success of new digital technologies may greatly depend on the services’ ability to collect and analyze *enough* personal information (Goldfarb and Tucker 2011, Johnson 2013b, Aziz and Telang 2015). The resulting products have enormous potential for offering better information flows, better choices, increased efficiency, and increased welfare. On the other hand, widespread unease and loss of trust in the market could result when providers store *too much* personal data (Miller and Tucker 2009; Acquisti et al. 2015, 2016).¹ Ultimately, too much data collection may also carry significant societal risks.²

In this paper, we analyze data from approximately 300,000 smartphone applications (“apps”) to inform the debate about optimal levels of privacy protection. Using data from Google’s Android Market, we study the extent to which private information resembles a second “means of payment” on both sides of the market for apps.³ Specifically, we use our data to analyze three research questions:

1. Do app developers request access to more sensitive data in exchange for lower prices?

2. Do users avoid installing apps with permissions that access privacy-sensitive data?

3. Are app users’ privacy concerns context dependent?

Our results provide, to our knowledge, the first large-scale empirical evidence of a money-for-privacy trade-off on both sides in the app market: app developers trade greater access to personal information for lower prices, and consumers choose between lower prices and greater privacy. More precisely, our results suggest that (1) app developers ask for more privacy-sensitive permissions if they offer a free app than if they offer a paid app, (2) consumers take this trade-off into account and show up to 6% less demand for apps that ask for privacy-sensitive permissions, and (3) the negative relationship between permissions and downloads depends on the app’s context. Factors such as trust, the targeted user group (e.g., mature users), and the sensitivity of the app’s context (e.g., in health-related apps) moderate the relationship’s strength.

Our data are relevant and informative for analyzing the money-for-privacy trade-off. The Android Market for apps has a large number and variety of products and is of high economic value. In most app categories,

users can choose among many alternatives. Privacy is salient, as users are shown the list of permissions requested by an app prior to installation. This policy creates a setting in which the money-for-privacy trade-off can be meaningfully studied, because we can discern 136 distinct permissions and record each app's permission requirements using data collected in 2012. We combine this information on each app's ability to collect personal information with a rich data set on 300,000 apps from Google's Android Market. The data cover publicly available app-specific information, including each app's number of installations, its price, and even each app's closest competitors. We collected data repeatedly in 2012 (over five months) and once in 2014. We augmented these data with information from [AppAnnie.com](#) and Apple's App Store to add further information about apps and contrast performance among operating systems.⁴

Our inference is based on a full cross section, on panel data sets, on "app siblings" consisting of a free and a paid version of the same app, and on a difference-in-differences-style analysis of demand for apps. Our results emerge consistently for both the supply and demand sides across these different data sets and in various specifications. The findings are robust to using alternative measures of demand, using other outcomes such as survival, measuring privacy-sensitive permissions in different ways, and considering subsegments of the market. They are also robust to an instrumental variables (IVs) approach that instruments for app prices or to leveraging our data on "app siblings"—a free version and a paid version of the same app—to identify and account for category-specific permissions (e.g., running apps that need to track location). A second source of identification comes from a cross-platform comparison between Android and Apple's iOS. We leverage the difference between Apple's and Google's notification policies regarding permissions and show that an app's download rank in Android is worse than in Apple's iOS if they require sensitive permissions. This finding corroborates our baseline conclusions.

Our paper follows the usual structure. Two online appendices contain additional supporting materials and cover the data in more detail.

2. Related Literature

By highlighting the money-for-privacy trade-off on both sides (demand and supply) of the market for mobile apps, we contribute to two streams of the literature: a large stream on the value of privacy and a growing stream on the mobile app market. Additionally, we hope to inform both policy makers and legal scholars concerned with privacy in mobile apps about the trade-offs in this sensitive market.

A central stream of the literature has studied the value of privacy both for the demand and the supply

side of markets. The key idea in the literature about the value of privacy is that consumers or users can be compensated for sharing information about their data or their activities with service providers and other parties. Whereas the large majority of the papers in the previous literature have investigated this idea in theoretical models, experiments, or surveys, we use a large observational data set. Moreover, we are unaware of any studies that use such data to analyze the role of privacy in the market for mobile apps, a market with unseen potential for collecting information about users. A major advantage of our setting is that the privacy permissions are unusually clearly communicated. Whereas past work found little demand for privacy, especially with permissive default settings (e.g., Gross and Acquisti 2005), the observed choices in the present setting reflect many deliberate choices by consumers to download apps with privacy being a component of that bundle. Thus, our setting adds an additional angle to the existing debate, because our inference is based on the users' actual choices in a new and highly privacy-sensitive market.

To motivate our research questions we can draw from a wealth of existing theoretical studies of privacy. Specifically, in most theoretical papers that model the demand side, the cost of privacy to consumers arises either as nuisance from too aggressively targeted ads that may trigger costly avoidance efforts (Hann et al. 2008, Johnson 2013b), or from a firm's ability to use knowledge about a user's preferences to price discriminate (Wathieu 2002, Conitzer et al. 2012).⁵ Existing literature identifies three key moderators of aggregate privacy concerns: (1) the composition of consumers, (2) trust-inspiring measures by firms, and (3) the nature and amount of the requested private information (Chellappa and Sin 2005, Chellappa et al. 2010, Acquisti et al. 2016). Similar factors arise in our empirical context and will be analyzed in Section 6.3, where we used the model by Chellappa et al. (2010) as a framework to guide our analysis of moderating factors.⁶ Note that the empirical literature also studied other sources of privacy concern than targeting and price discrimination. Such additional concerns arise from direct discrimination (e.g., in hiring, housing, healthcare or credit markets), or from fears of errors, improper access, or secondary uses (e.g., identity theft) of personal information (Acquisti et al. 2016). Compared with price discrimination and targeted advertising, consumers' concerns about direct discrimination or identity theft could be higher.⁷ Such concerns might be among the key factors that moderate the demand for privacy in our analysis. For example, we find evidence that consumers' concerns vary with an app's context, by showing that users are more cautious in especially sensitive settings, such as health-related apps.

More generally, users demand privacy and attribute value to not being targeted or "seen" in many settings

(Turow et al. 2009, Marthews and Tucker 2017). The demand for privacy was found to change over time and to depend on context or framing (Gross and Acquisti 2005, Goldfarb and Tucker 2012, Acquisti et al. 2013). Such findings are related to the so-called “privacy paradox” (see Norberg et al. 2007, Acquisti et al. 2016), which is a major topic of debate in the existing literature, and refers to the large difference between the estimates from observational versus experimental/survey-based evidence. We contribute an additional piece of evidence to this debate, because we use observational data on the app market, in which the use of personal data are a clearly communicated part of the bundle.

In other words, we contribute to the empirical literature on the value of privacy by measuring the managerially relevant magnitude of consumers’ demand for privacy, using large transaction-based data that reflect revealed preferences in the market for mobile apps. Existing empirical estimates of the value of private information in online markets (rather than mobile apps) are based on experimental and survey data. The estimated valuations ranged from zero to very large numbers (Grossklags and Acquisti 2007, Racherla et al. 2011, Tsai et al. 2011, Beresford et al. 2012, Carrascal et al. 2013). Our results suggest that the consumers’ revealed valuations for privacy in the 2012 mobile app market were so low that a business model which offers a privacy preserving app for small payment is not viable for most contexts or applications.

For the supply side—that is, for firms—a similarly rich literature has documented that user data are valuable. We extend this notion to the market for mobile apps. Data collection on the Internet is inexpensive, but it enhances the effectiveness of targeted advertisement, which affects revenues for content sites (Goldfarb and Tucker 2011; Tucker 2012, 2014; Aziz and Telang 2015).⁸ Moreover, digital services, such as apps or websites, can directly track their customers to learn who uses their service and how they use it (Mobasher et al. 2000, Hu et al. 2007). Thus, user data can help service providers to better understand their customer’s needs and how to improve their service (Tambe et al. 2012, Bertschek and Kesler 2017). The overwhelming insight from this literature is that user data can be a valuable resource for the firm. We add to it, by highlighting that developers offer their products at lower prices in exchange for additional data from their users.

Our paper also closes a gap in a second more recent stream of research, which used observational data to shed light on the market for mobile apps but did not focus on the role of privacy (Carare 2012, Bresnahan et al. 2014, Ghose and Han 2014, Liu et al. 2014, Yin et al. 2014, Askalidis 2015, Ershov 2016, Li et al. 2016, Chaudhari and Byers 2017, Hallen et al. 2017, Liu 2017).⁹ Their findings have increased our understanding of the app market but left room for studies that focus on

the role of private user data. The paper most similar to ours (Savage and Waldman 2015) uses survey-based data to estimate users’ average valuation of the data that users typically share with developers. The authors find a valuation of USD 4, which implies that offering more privacy could be a profitable market positioning for app developers.¹⁰ Our results based on observational data suggest that privacy-preserving app strategies may not be economically viable.¹¹ This difference both reaffirms the privacy paradox and shows the importance of considering observational data based on market transactions to understand the role of private user data in the market for mobile apps.¹²

To summarize, little is known about the role of private data in the market for mobile apps. We close this gap by highlighting the money-for-privacy trade-off on both sides (demand and supply) of this highly data-driven market. We show that users prefer and value apps with greater respect for their privacy, which confirms earlier research.¹³ In addition, we highlight app developers’ willingness to trade forgone revenue for personal user information. Thus, user data are a valuable resource for developers by which an emerging market can subsidize the creation of innovative products. We complete the picture by showing the context dependence of privacy concerns for mobile apps. Among these context-dependent factors are a developer’s outside reputation, the existence of privacy policies, and an app’s category (see Acquisti et al. 2015).

3. Background: Data Sharing in the Mobile App Industry

We begin this section by discussing the app market’s background and setup with regard to private data and then turn to the developers’ ability to resell an app’s data.

3.1. The App Market

In 2007, Apple introduced the iPhone. The device triggered a radical transformation of mobile communication in which screen-only smartphones replaced their predecessors. One of the main competitive advantages of the iPhone and its successors was a large app ecosystem. Apps allow users to tailor their devices to their needs by enabling multiple uses in addition to traditional phone calls and text-based apps. In 2008, the first smartphone using Google’s Android Operating System (Android OS) was released. Although users adopted initial versions slowly, Android gained popularity in 2010 and now dominates the market in most countries. According to the International Data Corporation (2015), the Android OS reached a market share of about 75% in 2012. In 2015, revenue from the mobile app store was nearly \$40 billion and is expected to reach \$100 billion in 2020 (App Annie 2016).

In addition to the Android OS, Google also introduced its own platform for the distribution of apps,

originally called Android Market but renamed Google Play Store in mid-2012. It serves as a distribution channel for apps, books, movies, music, and newspapers. By the end of 2012, the Google Play Store offered approximately 400,000 apps, and in 2015, the number increased to 1.5 million. Google distinguishes 30 categories of apps, such as communication, education, productivity, shopping, weather, and game apps. The last category, consisting of arcade and action, brain and puzzle, cards and casino, casual, racing, and sports games apps, makes up about 17% of our estimation sample.

A central feature of the Android app ecosystem is its permission system. This system is specific to the Android OS and provides a setting in which the money-for-privacy trade-off can be meaningfully studied. First, developers can choose among standardized blocks of information, so-called permissions, in which some enable access to a user's location, communication, browsing behavior, and so on. Before their installation, apps must declare which permissions they require and must request these from the users. More precisely, the system provides a list of permission names alongside a short explanation for each permission requested. Users must accept this list and explicitly acknowledge that they are granting these permissions to proceed with the installation. Alternatively, if they feel uncomfortable about the set of permissions requested, they can cancel the installation. Note that such explicit consumer consent for the set of permissions does not exist in Apple's iOS, in which this information remains implicit before installation. In essence, this procedure has been in place since 2012, despite the rapid growth of the Android Market.

In 2012, developers could choose among 136 predefined permissions.¹⁴ This large number illustrates the quantity and diversity of information app developers can potentially collect about app users. Online Appendix Figure A2 illustrates the way in which permissions were displayed in the Android Market in 2012.¹⁵ Since then, Google has introduced several small modifications in how permissions are displayed to the user. Before 2014, the list of permissions showed permission names next to short explanations of the permissions. Since 2014, the system shows only the names of aggregated permission *groups* (although users can open a more detailed dialogue for each permission group). Still, users must approve of the permission list before proceeding with installation.¹⁶

3.2. Monetization and Trading of App Data

Developers can monetize their apps via four main channels. According to AppBrain (2016), about 20% of the apps in the Google Play Store are paid apps, whereas the remaining 80% of apps are free.¹⁷ Alternative revenue channels are in-app advertising, in-app purchases, and data trade. The size of these alternative revenue channels has been relatively stable since 2012

except for in-app purchases, which were introduced shortly before our period of observation.¹⁸ In 2012, when we collected our data, the "freemium" model based on in-app purchases barely existed. Since then, a marked increase has occurred in the use of this model, in which users may install the apps without paying but must pay to unlock useful additional functions. The two other channels, in-app advertising and data trade, were already common.

Data sharing for monetization purposes is common in the mobile app industry, as in many online markets (see, e.g., Woodcock 2017 and references therein). Christl and Spiekermann (2016) and Organisation for Economic Co-operation and Development (2013) survey several studies that show that apps commonly shared data with third parties.¹⁹ They also provide a brief discussion of the data-sharing business model and the three main channels that developers have to exploit app usage data to generate revenues: (1) The most important channel is using information about users to sell targeted advertising.²⁰ This channel has arguably been the powerhouse of the market for free apps, representing a sizable industry.²¹ The value in this industry is generated from collecting and aggregating individuals' digital traces from various sources (e.g., an app) to profile users' location (Fong et al. 2015, Dubé et al. 2017) and behavior for targeted advertising.²² Yet, although this type of information brokerage potentially offers greater efficiency in matching the ads, sharing the data implies that the app users' information is passed on, and their safety depends solely on the integrity of the developer's advertising partner. (2) Another prominent channel of data flows arises when developers trade their data for valuable third-party services. A good example of such services is app analytics. App analytics help developers gain insight on who uses their app when and where, along with which other processes on the phone are simultaneously being used. Combined with the developer's own knowledge of the user's in-app behavior, this can be a useful input for improving the app or, again, for advertising purposes. As with third-party advertising, the secondary usage of the user data depends on the analytics site's own policy and integrity. (3) Finally, app developers can trade their data with third parties in several other common ways. For example, they can exchange their data for direct monetary benefit. Such data can be purchased by known industry analysts such as Alexa.com or by lesser-known data aggregators. As with third-party libraries, the secondary usage of the user data depends on the analytics site's policy and integrity.

4. Data and Descriptive Evidence

We first describe our main data set and then turn to our two alternative data sets (see Section 4.2). The descriptive analysis in Section 4.3 illustrates two key

findings about the money-for-privacy trade-off in the market for mobile apps.

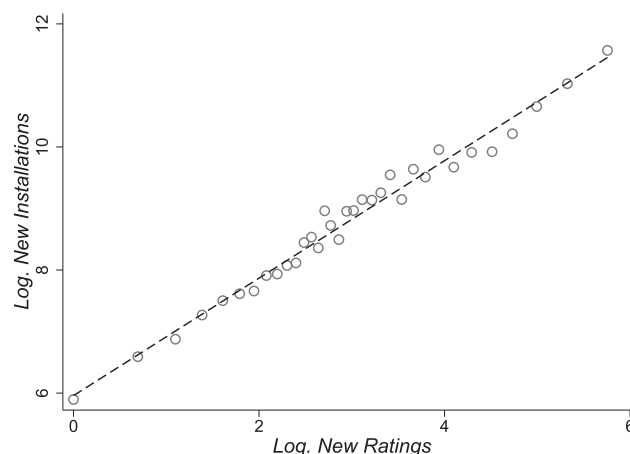
4.1. Main Data and Variables

For our main data set, we extracted all publicly available information on nearly all the products available in the Google Play Store in 2012 (approximately 300,000 apps). We collected the data on a monthly basis from April to September 2012. Online Appendix Figure A1 shows the design of Google's app store in 2012, which corresponds to the information we were able to collect. Using this information, we created two main data sets: (1) a cross section of all apps available in April 2012 and (2) a balanced monthly panel of apps that were available from April 2012 until September 2012 and had within-variation in the number of privacy-sensitive permissions. In addition, we collected a cross section of data in 2014 to analyze long-term outcomes. Also, for some of the robustness checks, we used additional data sources, such as [Alexa.com](#), Amazon Mechanical Turk, [PrivacyGrade.org](#), and [AppAnnie.com](#).

Given our research questions, we need three main types of information: a price measure, a demand measure, and a measure of an app's ability to collect private information about users. In the following, we introduce and discuss each of these measures as well as the core control variables.

4.1.1. Main Outcome Variables. Our first main outcome is the *price* of apps. We directly observe price (in euros) in the data. We measure (1) the decision to provide a free app or a paid app as a dummy ($D_{\text{Paid}} = 1$ if the app is paid) and (2) the price if a paid app is offered ($\ln(\text{price})$). Our second main outcome is *demand*, which we observe with two measures. Our first measure of app demand is the monthly change in the number of ratings ($\Delta \text{Ratings}$). Alternatively, we use the monthly change in the number of installations ($\Delta \text{Installations}$). Both proxies measure the number of new customers who consider the benefits of installing an app greater than the cost of paying the product's price or the associated loss of privacy. The two demand measures are highly correlated (see Figure 1 and Online Appendix A.2). The high correlation is not surprising given that users have to install an app in order to rate it. Our preferred demand variable is the change in the number of ratings, because we observe the exact number of ratings, whereas the number of installations is observed only in discrete step-size form with 17 levels (e.g., 11–50, 51–100, 101–500 installations), which leads to low intertemporal variation in a panel specification.²³ It is well known that both ratings and installations are not without drawbacks, for example, because both can be purchased by developers to promote their app (Li et al. 2016) and only a small fraction of users who install an app might actually use or rate it (Hu et al. 2009).²⁴

Figure 1. Relationship Between New Installations and New Ratings



Notes. The figure displays the relationship between apps' log number of new installations and their log number of new ratings. A linear trend is added. The data used are an aggregated version of the cross-sectional data set used in the estimation sections.

Hence, we show the robustness of our findings with respect to the choice and definition of our demand measure in Section 6.4. There and in our main demand specifications, we use several alternative demand measures based on ratings, installations, rankings, and alternative time windows.²⁵ The industry's most relevant measure for an app's demand or usage would be daily active users (DAU), but this measure is not available, because it is only known by an app's developer team.

4.1.2. Identifying the Privacy Sensitivity of Apps. To measure an app's ability to collect private information, we take advantage of the fact that, as described earlier, Google provides precise details on the permissions an app uses. This feature allows us to understand exactly what private information an app can collect about the app user. Among the 136 permissions available to app developers, some can be considered innocuous with respect to the privacy of the user, whereas others grant an app access to sensitive information. On the basis of an extensive literature review (Dini et al. 2012, Jeon et al. 2012, Sarma et al. 2012, Mylonas et al. 2014, Taylor and Martinovic 2016), we were able to identify a total of 25 privacy-sensitive permissions, which allow apps to access information about users' ID, location, profile, or communication (see Online Appendix Table A1). These potentially problematic permissions include "fine (GPS) location," which allows access to the user's current location; "read browser data," which grants access to a user's browsing history and their bookmarks; and "read contact data," which allows an app to access the user's contact data. From this classification, we construct our main variable of interest (D_{Privacy}), which is a dummy equal to 1 if an app uses at least one of the 25 privacy-sensitive permissions and 0 otherwise. To capture the

intensity of an app's ability to collect private information, we additionally use the number of privacy-sensitive permissions in an app (*#Privacy*). In addition, we evaluate our findings when using other, alternative privacy measures in Section 6.4. These measures include (1) a category-specific modification of our baseline measure, (2) a classification based on Google's assessment of problematic permissions, (3) a classification derived by hiring 400 classifiers at Amazon Mechanical Turk, and (4) a classification to account for the fact that apps require internet access to transmit user information. Online Appendix Tables A1 and A2 summarize all classifications applied and describe all privacy-sensitive permission. In addition, these tables show the details of how we grouped the privacy-sensitive permissions into four subgroups: location-, profile-, communication-, and ID-specific permissions.

4.1.3. Control Variables. In addition to our main variables, we also observe a rich set of app-specific characteristics relevant for explaining app supply and demand: the app category, the average rating, the code size, the required version of Android OS, developer-specific information (name of developer, number of its other apps, top developer status, etc.), and the app's description (length, number of screenshots, video) and permissions related to internet access (D_{Internet}) and advertising (D_{Ads}).²⁶ Also, we use the section "users who viewed this also viewed" to identify app-specific competitors and to construct three additional control variables: the average price of competing apps, competitors' average number of installations, and the average rating of competing apps. Finally, to control for an app's functionality unrelated to the ability to collect private information, we include in all specifications the number of unproblematic permissions to which an app has access (*#CleanPerm*). For a list and definition of our key variables, see Online Appendix Table A2.

4.2. Two Alternative Data Sets

In addition to our two main data sets (cross section and balanced panel from 2012), we additionally use a set of app pairs and a set of apps for which we could retrieve app download rankings from 2012 in both the Google Play Store and Apple's App Store. We use these two alternative data sets for robustness checks that control for the unobserved heterogeneity of apps. We also use the second one to show the close relationship between app download rankings and our demand measure.

4.2.1. App Pairs. App developers often offer two versions of the same app, a free version and a paid version. The paid version typically offers some advantages over the free version: it may offer additional functionality, contain less advertising, or require fewer (privacy-sensitive) permissions.²⁷ We identified pairs within our

cross-sectional data set, which consist of app siblings with the same functionality.²⁸ To do so, we used a two-to-three-step procedure. First, we used a word-processing algorithm that identified app pairs with the same name except for the addition of one of the following: "free," "paid," "lite," "full," "demo," "pro," "premium," "donate," "trial," and "plus" (which results in 7,211 app pairs). Second, within this broad set of pairs, we identified those for which the code size of the paid version is not larger than that of the free version and for which the description of both apps is more or less the same length.²⁹ This gives us a sample of 1,999 siblings. Third, to identify an even more rigorously matched subset of app "twins" we *manually* checked the description of the remaining app siblings and kept those whose description indicated no difference in functionality. The versions in this last set of twins should thus differ only in price, permissions, and possibly the existence of ads. This most restrictively matched sample consists of 354 pairs and even excludes app siblings with identical descriptions if the text mentioned any differences between free and paid versions.

4.2.2. Android and iOS App Ranking. Many apps are offered for both Google's Android OS and Apple's iOS. This fact allows us to compare the relative success of apps in both systems and the extent to which it depends on their use of privacy-sensitive permissions. We used AppAnnie.com to retrospectively collect (in 2016) the apps' rankings from 2012 for both OS. Unfortunately, in 2016, we found only a small number of 192 (mostly successful) apps for which the app ranking from 2012 was available for both OS. Nevertheless, the data set allows a rigorous comparison across platforms. Unlike in our main data sets, which contain global success measures, such as the worldwide number of installations, rankings are available only at a country-specific level. Thus, we collected and aggregated rankings for seven large markets: Germany, India, Japan, South Korea, Russia, the United Kingdom, and the United States. For these countries, we have rankings for April and September 2012. For free apps we use the overall free app rank, and for paid apps we use the overall paid ranking.³⁰ From the collected country-, time-, and OS-specific ranks, we compute average OS-specific values for each app. We use the order of these average values to construct our own OS-specific in-sample rankings, which are designed to ensure that the values are comparable across OS and limit the potential effect of outliers. Using these standardized in-sample rankings, we construct two measures of relative app success on the two operating systems ($\Delta \text{Rank}^{\text{iOS-And}}$): We compute (1) the difference between rankings for Android OS and iOS based on the average OS-specific rankings that we were able to collect for the two versions of the app and

(2) the same difference but based only on the ranking information of the largest market (i.e., the United States).³¹ This data set contains up to 192 apps observed in both OS.

4.3. Descriptive Evidence

Before providing descriptive evidence on two key results of this paper, we discuss the summary statistics of the variables in our main data sets.

4.3.1. Summary Statistics. Table 1 provides an overview of the most relevant variables in our main data sets. For each data set, the left column shows averages for free apps, and the right column shows those for paid apps. Columns (1) and (2) show permission use in the entire cross section of apps, consisting of 177,193 observations.³² Free apps are installed more often and have a lower average rating. Crucially, an average free app

is more likely to use potentially privacy-sensitive permissions and will require more such permissions than an average paid app. For example, free apps use on average 1.19 privacy-sensitive permissions, whereas paid apps use only 0.52 such permissions. Similarly, 50% of the free apps have at least one privacy-sensitive permission, whereas only 28% of the paid apps use such permissions. This relation does not change when we compare alternative groups of privacy-sensitive permissions, and this pattern also holds in our three other data sets: the panel data set of apps that have within-variation in the number of privacy-sensitive permissions over time (columns (3) and (4)), one of our sets of matched app siblings (columns (5) and (6)), and the set of apps for which we were able to retrieve app rankings from 2012 for both Android OS and iOS. Despite covering very different subsets of apps, all four data sets show consistent patterns with respect to the

Table 1. Summary Statistics of the Main Data Sets

	Cross section		Panel		Siblings		Android vs. iOS	
	Free (1)	Paid (2)	Free (3)	Paid (4)	Free (5)	Paid (6)	Free (7)	Paid (8)
Outcome measures								
Δ Ratings	9.23	2.31	89.09	34.19	67.33	6.08	27,107.85	807.74
Δ Installations	3,178	105	16,561	743	8,213	93	1.5e+06	61,333
Ratings	158.54	44.50	1,083.25	372.03	623.53	81.05	1.7e+05	5,096.88
Installations	43,376	1,936	1.9e+05	11,719	1.3e+05	2,740	2.7e+07	2.7e+05
Price	0.00	2.01	0.00	4.41	0.00	1.37	0.00	3.04
Permissions								
#TotalPerm	4.10	2.32	8.08	6.67	3.78	3.10	11.17	5.71
DPrivacy	0.50	0.28	0.87	0.76	0.49	0.40	0.95	0.60
#Privacy	1.19	0.52	2.82	1.88	0.93	0.70	3.40	1.03
#CleanPerm	2.92	1.80	5.26	4.79	2.85	2.40	7.77	4.68
DPrivCatSpec	0.22	0.09	0.52	0.27	0.14	0.08	0.38	0.12
DMTurkEP2	0.12	0.08	0.22	0.27	0.12	0.12	0.39	0.11
DGoogle	0.35	0.16	0.66	0.49	0.25	0.21	0.64	0.20
DSarmaetal	0.47	0.25	0.85	0.70	0.46	0.35	0.90	0.52
DID	0.32	0.14	0.69	0.47	0.30	0.22	0.76	0.42
DLocation	0.29	0.10	0.61	0.31	0.21	0.12	0.46	0.17
DCommunication	0.06	0.05	0.14	0.18	0.06	0.06	0.21	0.06
DProfile	0.21	0.11	0.44	0.41	0.19	0.18	0.57	0.22
DInternet	0.81	0.44	0.98	0.85	0.82	0.56	1.00	0.86
DAds	0.55	0.21	0.89	0.63	0.56	0.31	0.91	0.60
Control variables								
Average Rating	3.91	3.99	3.98	4.07	3.79	4.14	4.38	4.21
Size (in KB)	2,192	3,928	3,684	8,886	3,207	3,039	10,324	13,034
Length Description	730	996	1,004	1,612	877	874	1,717	2,019
Number Screenshots	3.36	3.69	4.08	5.25	3.73	3.87	5.56	5.17
Dummy: Video	0.10	0.12	0.12	0.24	0.17	0.17	0.47	0.59
Dummy: Top-Dev.	0.00	0.01	0.01	0.04	0.01	0.01	0.61	0.43
Apps by Developer	92	127	63	34	14	14	10	8
Average Installations of Developer	74,235	58,693	1.4e+05	1.6e+05	57,843	91,749	1.1e+07	2.5e+06
Observations	128,921	48,272	30,131	2,964	1,999	1,999	94	98

Notes. The table provides an overview over the most important variables, and shows the corresponding descriptive statistics for the four main datasets in this paper. For each data set we show two columns, where the first column shows averages for free apps and the second column for paid apps. The first two columns show the summary statistics for the entire cross section. The second pair of columns ((3) and (4)) shows those for our panel data set consisting of apps having varied their use of privacy-sensitive permissions at least once between April and September 2012. Columns (5) and (6) use the data set consisting of app pairs where the paid version of the app has the same or a smaller code size and where both apps have more or less the same description length. Columns (7) and (8) show statistics for those apps for which we were able to collect 2012 information on app ranks both in the Play Store and in the iOS store.

differences in the permission use by free and paid apps: No matter how we look at the data, free apps always request more privacy-sensitive permissions than do paid apps. This holds for essentially all criteria we analyzed, and this pattern is found even when we match pairs of apps that come from the same developer and have the same functionality. We provide additional information in Online Appendix Table A2 and summary statistics for the cross-sectional data (Online Appendix Table A3) and pairs (Online Appendix Table B1).

Below, we document two key results about the supply side. First, some mainly free apps request permissions that are not necessary for their functionality. We conjecture that these unnecessary permissions are used for monetization. Second, privacy-sensitive permissions are more likely to appear in free apps than in paid apps. This indicates a negative relationship between price and permission use. Both results suggest a trade-off between price and privacy. Consumers can choose between apps that are cheaper but request more privacy-sensitive information and apps that are more expensive but more privacy respecting.

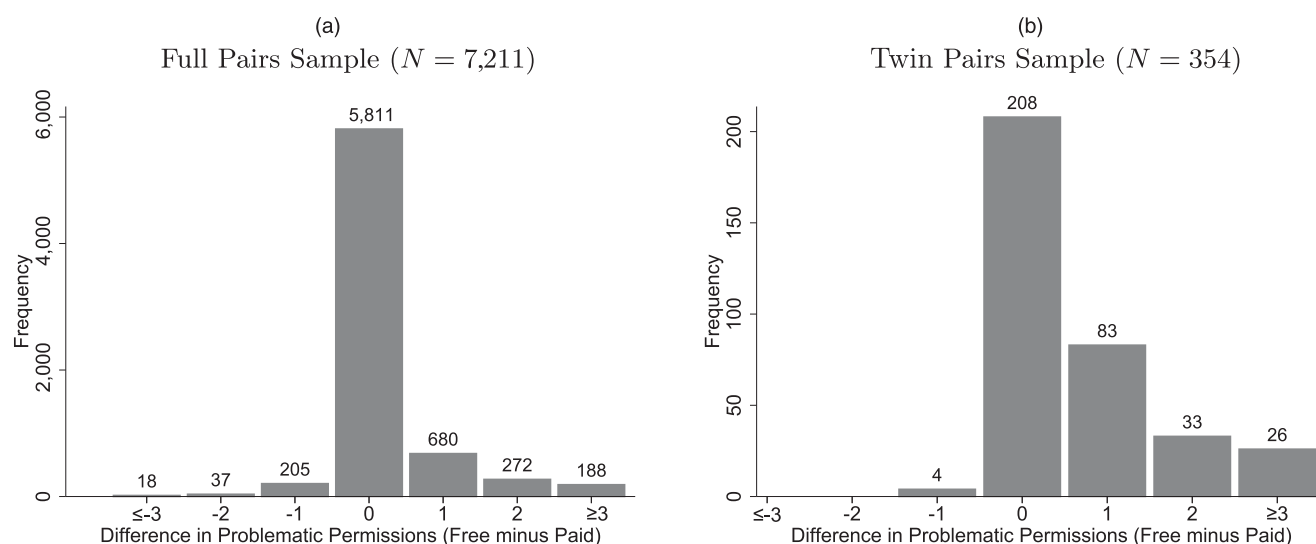
4.3.2. Unnecessary Permissions. We use both the full sample of app pairs and the sample of selected app twins to study whether apps request permissions without needing them for their functionality. In both

data sets, the paid version serves as a technological reference, because the paid version can be safely assumed to provide the same or even more functionality than the free one.³³ Hence, any permission that is present in the free version but not in the paid sibling is unnecessary in terms of functionality and can be expected to be related to monetization.

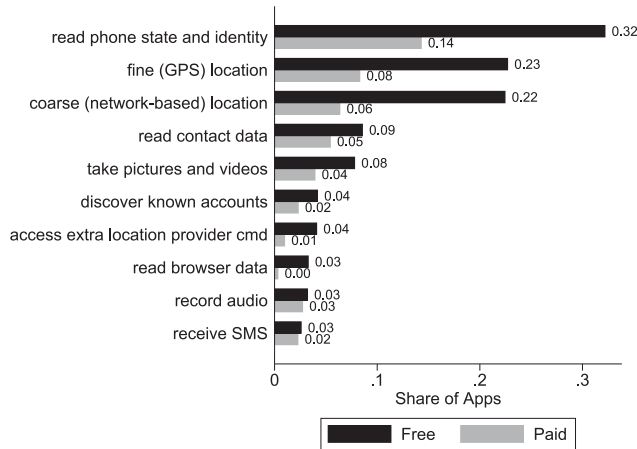
Figure 2 compares the number of privacy-sensitive permissions in a pair's free and paid versions. Panel (a) shows how frequently we observed a given difference in permissions for the full pairs sample. A frequency of 280 for a value of 2 means that we find 280 pairs for which the free version requested two permissions more than the paid version. Panel (b) shows the same comparison for the selected sample of app twins, for which we ensured that functionality does not differ between apps. Across both samples, we find more pairs in which the free version requires more privacy-sensitive permissions than the paid one. Particularly in panel (b), the paid version hardly ever requests more permissions than its free counterpart. We conclude that several of the free versions request unnecessary, privacy-sensitive permissions that are not required for their functionality.

4.3.3. Price and Permissions. When we analyze all apps, we see that developers, who charge zero prices, are more likely to request more privacy-sensitive permissions. Figure 3 shows the 10 most frequently used

Figure 2. Difference in Privacy-Sensitive Permissions Between Free and Paid App Siblings



Notes. Both figures compare free and paid versions of an app pair. On the x axis we show the differences in the number of privacy-sensitive permissions when comparing the pair's free and paid versions (equal to the number of privacy-sensitive permissions of the free app minus the number of privacy-sensitive permissions of the paid twin). The y axis shows the frequency with which each difference occurs. Panel (a) shows those frequencies for the full pairs sample, whereas panel (b) shows them for the selected sample of app twins where we ensure that the functionality between apps does not differ. In most cases both versions request the same number of permissions (80% of all pairs and 59% of the pairs with matched functionality). Whereas the difference in average permission usage is less pronounced when analyzing all pairs, the free app is much more likely to request sensitive permissions once we condition on observing a difference. This pattern emerges for both pair samples and is strongest on the pairs with no discernable difference in functionality (1,151 vs. 267 or 80% vs. 20% overall and 142 vs. 4 or 97% vs. 3% on pairs with matched functionality).

Figure 3. Frequency of Top 10 Privacy-Sensitive Permissions—Free and Paid

Notes. This diagram shows how often privacy-sensitive permissions are used, both by free and by paid apps. The majority of apps do not use these permissions. However, given that such permissions are used, free apps use them more frequently than do paid apps. The data used are equal to the cross-sectional data set used in the estimation sections.

privacy-sensitive permissions and contrasts the share of free and paid apps that request them. All privacy-sensitive permissions are more common in free apps. Developers choose between either offering their apps free with additional privacy-sensitive permissions or charging a fee but including fewer privacy-sensitive permissions.

The goal of the subsequent sections is to provide rigorous econometric evidence of the trade-off that is implied by the key findings we just presented.

5. Empirical Framework

This section discusses our empirical framework. The first subsection presents an overview of our estimation approach for the supply side. Subsequently, we discuss estimation of the demand side.

5.1. Supply-Side Estimation

We run two types of correlational supply-side regressions to provide insights on the role of private information for app developers. First, we examine the choice of the business model. Specifically, we analyze how the developer's decision to offer a free app or a paid app is related to the app's use of privacy-sensitive permissions. Our estimation equation is

$$D_i^{\text{Paid}} = \alpha + \beta D_i^{\text{Privacy}} + \theta X_i + \varepsilon_i, \quad (1)$$

where D_i^{Paid} is a dummy equal to 1 if the app i is paid, and ε_i is the error term. The dummy D_i^{Privacy} is equal to 1 if an app requires at least one privacy-sensitive permission. A negative β indicates that free apps are more likely to request privacy-sensitive permissions.³⁴

The control variables X_i are the number of unproblematic permissions, a dummy for ad-related permissions, a dummy for internet access, dummies for the app category and the maturity level (e.g., "recommended for users aged 13+"), the average rating (in logs), the code size (in logs), the app version, the length of the app description (in logs), the number of screenshots, a dummy for the existence of a video, a top-developer dummy, the number of apps by the developer (in logs), the average number of installations of the developer's other apps (in logs), the minimum and the maximum compatible Android OS version, and information about all of the competing apps' characteristics.

In a second specification we restrict our sample to paid apps and study how price levels are related to the use of privacy-sensitive permissions. The estimation equation is

$$\ln \text{Price}_i = \alpha + \beta D_i^{\text{Privacy}} + \theta X_i + \varepsilon_i, \quad (2)$$

where the dependent variable is the log-standardized price of an app. A negative coefficient β indicates a lower price for apps that require privacy-sensitive permissions. We stress that these cross-sectional regressions do not account for unobserved heterogeneity.

5.1.1. Panel Data Analysis. We apply two main strategies to address the concerns over unobserved heterogeneity and to validate our results on alternative data structures. In our first approach, we use a panel specification to control for unobserved heterogeneity. We include an app fixed effect (α_i) and estimate our model based on within-app variation. We combined the cross sections from April and September 2012 to form a panel covering our observation period of five months. Analyzing the within-app variation focuses on changes in price or the app's business model and allows us to control for unobserved heterogeneity. However, this approach is limited by the small number of apps that changed price or switched business model during our sample period. For the business model, we estimate the following fixed effects model:

$$D_{it}^{\text{Paid}} = \alpha_i + \beta D_{it}^{\text{Privacy}} + \theta X_{it} + \varepsilon_{it}, \quad (3)$$

where α_i captures unobserved heterogeneity of app i , and t represents the month. We estimate the analogous relationship for the price of apps (provided they request payment) and use the waves from April and September 2012 for both specifications.

5.1.2. App Pairs. Our second main strategy to deal with unobserved heterogeneity exploits variation within pairs of apps, in which the free and the paid versions of an app differ mainly in price and the number of permissions they use. These app siblings shed light on the

money-for-privacy trade-off, as perceived by developers. As explained in Section 4.2, this is because the paid version serves as a technological reference that allows us to identify permissions that likely serve purposes other than facilitating the app's core services. The regression equation is

$$D_{pi}^{\text{Paid}} = \alpha_p + \beta D_{pi}^{\text{Privacy}} + \theta X_{pi} + \varepsilon_{pi},$$

where α_p controls for unobserved heterogeneity of app pair p . This framework is valid if we can ensure that the two siblings have identical functionality—hence the need to identify app pairs without any discernible difference in functionality, which differ only in permissions and price.³⁵ We can run this regression only for the app's business model choice, because one of the siblings is free by definition.

5.2. Demand-Side Estimation

Our baseline specification to analyze the relationship between app demand and permissions is based on the cross section sample. We model demand as a function of its permissions, its price $Price_i$, and the same set of control variables X_i as in the supply-side specifications. We estimate the following baseline model:

$$\ln Demand_i = \alpha + \beta D_i^{\text{Privacy}} + \gamma \ln Price_i + \theta X_i + \varepsilon_i. \quad (4)$$

In the main specification, we approximate $Demand_i$ for app i by its change in the number of ratings in that month ($\Delta Ratings$), and in our robustness checks, we use installation-based measures. Again, β is the coefficient of interest, indicating that the use of a certain permission comes with a $\beta \times 100\%$ change in demand. Prices are in logs, so that γ can be interpreted as own-price elasticity.³⁶

Because ordinary least squares (OLS) is sensitive to unobserved heterogeneity, we test our results using (1) a two-stage least squares (2SLS) model, (2) a panel approach, (3) a difference-in-differences-style analysis between Android OS and iOS, and (4) an extensive series of robustness checks, which we present in the following section.

5.2.1. Endogeneity of Prices, Permissions, and Exit. Both permissions and price are strategic choices for the developer. Hence, the estimated demand coefficients might suffer from endogeneity bias. For monetary prices, endogeneity is well understood and usually leads to an upward bias in the estimated price coefficients (Wooldridge 2010). We address this issue using standard instruments for price, such as cost shifters including the code size and the number of apps a developer offers (Berry et al. 1995, Hausman 1996, Nevo 2000).³⁷

Using 2SLS techniques, we test whether our coefficients of interest (privacy-sensitive permissions) are

affected by the potential endogeneity in monetary prices. We show these results, which confirm our baseline findings, in Online Appendix Table A10.

The second source of potential endogeneity is permissions. If developers charge a higher “permission price” for unobserved high quality, this could introduce a positive correlation between permissions and the error term. The resulting estimation bias would be in the same upward direction as for the monetary price. We attempt to reduce the effect of unobserved quality using panel estimation techniques and our difference-in-differences-style approach.³⁸ Any remaining correlation would lead to an upward bias in the regression coefficients, and a truly negative coefficient would be biased toward zero. Without an instrument, we can only provide a lower-bound estimate of the effect of permissions on demand. Although it is hard to find the ideal instrument for this source of endogeneity, we can use developer's and competitors' behavior as an instrument. Specifically, we use the same developer's share of apps with privacy-sensitive permissions (excluding the focal app) as our first set of IVs. Alternatively, we use the share of the competing apps that use privacy-sensitive permissions as a second IV. Although both approaches suffer from limitations, they are good predictors of the app's permission choices and arguably do not directly depend on the app's success. Again, the results are shown in Online Appendix Table A10 of Appendix A.1.

A third source of endogeneity could be a survivor bias that could affect our cross section specifications. This is because our cross section consists only of survivors (until 2012), and the likelihood of app survival might depend on an app's use of permissions. If app survival is negatively related to permission use, (e.g., through app demand), we would observe only apps that are relatively successful such that a potentially negative permission effect on demand would again be underestimated. To study whether the use of privacy-sensitive permissions is related to apps' likelihood of survival, we correct for the potential bias from this selection process. We provide results based on Heckman selection models (Heckman 1979), which confirm our baseline findings, in Online Appendix Table A10.

Our remaining strategies to deal with the threats to identification are based on using different data structures that aim to eliminate unobserved heterogeneity.

5.2.2. Panel Data Analysis. In our first approach to tackling unobserved heterogeneity, we use an app fixed effect (α_i) to estimate a panel regression model. We thus only exploit within-app variation to address the concern that we cannot observe all heterogeneity between apps in the cross-sectional analysis, despite our rich set of control variables. In particular, unmeasured quality and functionality of apps could be positively correlated with

both app demand and permission use (resulting in an upward bias). To address this concern, we estimate the following fixed-effects specification:

$$\ln Demand_{it} = \alpha_i + \beta D_{it}^{\text{Privacy}} + \gamma \ln Price_{it} + \theta X_{it} + \varepsilon_{it}, \quad (5)$$

where $Demand_{it}$ is measured as the monthly change in the number of ratings, and the interpretation of the coefficient of interest, β , is the same as for the cross-sectional results. A negative coefficient would indicate that increased permission use comes with a subsequent decrease in demand.

5.2.3. Difference-in-Differences–Style Analysis. Our second strategy to tackle the challenge of unobserved heterogeneity between apps is a difference-in-differences–style analysis. Specifically, we compare the relative performance of apps with and without privacy-sensitive permissions that are available on both Android OS and iOS and are successful enough that we could obtain their 2012 download ranking retrospectively. We run the following regression:

$$\Delta Rank_i^{\text{iOS-And}} = \alpha + \beta D_i^{\text{Privacy}} + \gamma \#CleanPerm_i + \varepsilon_i,$$

with $\Delta Rank_i^{\text{iOS-And}}$ being defined as the difference between app i 's Apple and Android standardized in-sample download ranks ($iOSRank_i - AndroidRank_i$).³⁹ A negative β -coefficient indicates that apps that use privacy-sensitive permissions have a lower download rank in Apple's OS compared with Android's OS (i.e., they are downloaded relatively more often in iOS than in Android OS). This approach exploits the fact that the Apple platform is less explicit about app permissions when users download the apps than the Android counterpart. Therefore, Apple users may be less aware than Android users about the private information they are sharing through the app. We thus obtain two points of observations from each platform and can difference out the innate app characteristics in this comparison. This allows us to measure the effect of app permissions more precisely.⁴⁰ The downside of this analysis is the small and selective sample, which consists of only 192 apps in the top segment. However, although this analysis can only complement our main analysis of the demand side, it can do so with a relevant set of widely used apps and an especially clean design.

6. Results

We first analyze whether cheaper apps request more privacy-sensitive permissions. Second, we present our results on the demand side. Third, we analyze circumstantial factors (such as the reputation of app developers), robustness, and the implied valuations of private data.

6.1. Money vs. Privacy on the Supply Side

Table 2 shows descriptive regressions that relate the supply side's pricing choices to the use of privacy-sensitive permissions. The two outcomes of interest are the app's business model (columns (1)–(6)) and the price charged, given that it was positive (columns (7)–(10)). All these regressions control for a large set of variables that could drive app supply. These controls include the number of unproblematic permissions and permissions that are related to ads and internet access. Columns (1)–(6) analyze the developer's decision to offer a free app or a paid app. The dependent variable is a dummy that equals 1 if users have to pay a positive price for downloading the app. Columns (1) and (2) analyze the cross section; columns (3) and (4) use the panel of apps that change their business model at least once over the five-month sample period. Hence, we use only the first and the last waves of our panel in these regressions. Columns (5) and (6) use the app pairs. In the cross section, we find that apps that require privacy-sensitive permissions are 3.3% more likely to be free. Also, when we use the number of permissions to measure the presence of privacy sensitive permissions, we find that apps are more likely to be free when they have one or more sensitive permissions.⁴¹ Thus, the correlational results confirm the descriptive evidence showing that "a price comes with fewer privacy-sensitive permissions." This applies to both the presence of any privacy-sensitive permissions (column (1)) and the number of sensitive permissions (column (2)). The panel regressions in columns (3) and (4) highlight two patterns. First, fewer than 1 in 1,000 apps switch their business model and the number of permissions they use over a period of five months. This low percentage suggests that the choice of the pricing model is a strategic decision, which is not easily adjusted. Second, when using the dummy that indicates a switch between models, we find that moving from paid to free coincides with an insignificant increase in the likelihood that the app requests privacy-sensitive and ad-related permissions. However, analyzing the number of privacy-sensitive permissions (column (4)), we see that more such permissions come with a weakly significantly lower likelihood that an app is paid. Our analysis of the app pairs (columns (5) and (6)) confirms the cross-sectional patterns even more strongly. If one of the siblings uses sensitive permissions, it is 15.6% more likely to be the free version, which again is confirmed if we use the number of permissions rather than the dummy variable.

In columns (7)–(10) we analyze the determinants of paid apps' price level. Columns (7) and (8) show cross section results, and columns (9) and (10) use the panel data. First, note that only approximately 47,000 apps (26.75%) have a positive price, and of those, only 211 apps (less than 1%) had any variation in price and the

Table 2. Main Supply-Side Results

	Business model choice (D_{Paid})						Price choice (log price)			
	Cross section		Panel		Pairs		Cross section		Panel	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
D_{Privacy}	−0.033*** (0.002)		−0.104 (0.261)		−0.156*** (0.042)		0.012 (0.008)		−0.119* (0.070)	
#Privacy		−0.024*** (0.001)		−0.150* (0.086)		−0.137*** (0.025)		−0.005 (0.004)		−0.060** (0.025)
#CleanPerm	0.005*** (0.001)	0.011*** (0.001)	−0.003 (0.024)	0.035 (0.030)	0.204*** (0.023)	0.226*** (0.025)	0.018*** (0.002)	0.020*** (0.002)	0.048** (0.020)	0.068*** (0.023)
D_{Internet}	−0.219*** (0.003)	−0.225*** (0.003)	−0.627** (0.290)	−0.582** (0.246)	−0.496*** (0.037)	−0.517*** (0.038)	0.067*** (0.007)	0.068*** (0.007)	0.004 (0.085)	−0.015 (0.083)
D_{Ads}	−0.119*** (0.002)	−0.124*** (0.002)	−0.347 (0.301)	−0.165 (0.250)	−0.535*** (0.036)	−0.547*** (0.036)	0.014 (0.009)	0.014 (0.009)	−0.003 (0.119)	−0.039 (0.112)
Constant	0.094*** (0.035)	0.090** (0.035)	10.787*** (2.840)	9.913*** (2.754)	3.495** (1.533)	3.291** (1.520)	0.060 (0.106)	0.065 (0.106)	−2.943*** (0.870)	−3.558*** (0.835)
Category	Yes	Yes	Yes	Yes	No	No	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	176,000	176,000	96	96	3,998	3,998	47,079	47,079	422	422
No. of groups			48	48	1,999	1,999			211	211
Mean of dependent variable	0.27	0.27	0.50	0.50	0.50	0.50	0.20	0.20	0.53	0.53
SD of dependent variable	0.44	0.44	0.50	0.50	0.50	0.50	0.60	0.60	0.62	0.62
Adjusted R^2	0.320	0.322	0.963	0.971	0.618	0.622	0.223	0.223	0.177	0.183

Notes. The table shows the relationship between privacy-sensitive permissions and the strategic choices of app developers: the choice of the business model in columns (1)–(6) and the price choice in columns (7)–(10). In columns (1)–(6) the dependent variable D_{Paid} measures the developer's decision to offer an app for money or for free. It takes the value 1 if users have to pay to download the app. Columns (1) and (2) show descriptive regressions based on the cross section of data, where the independent variable of interest is (1) an indicator for one or more privacy-sensitive permissions (D_{Privacy} in column (1)) or (2) the number of privacy-sensitive permissions (#Privacy in column (2)). Columns (3) and (4) show panel fixed-effects regressions where we restrict the sample to such apps that changed both the number of privacy-sensitive permissions and the business model at least once between April and September 2012. Here, we use only the first and the last waves of our data to maximize the within-variation. Columns (5) and (6) use data on app pairs where the paid version of the app has the same or a smaller code size and where both apps have more or less the same description length. Columns (7)–(10) show the results for price-level choices (of paid apps). The dependent variable is the app's price (in logs). Columns (7) and (8) show cross-sectional regressions, and columns (9) and (10) show panel fixed-effects regressions based on the first and last waves of our data, where we restrict the sample to such apps that change both the number of privacy-sensitive permissions and their price at least once. In all specifications we drop outliers with respect to app prices (i.e., apps with prices above eight euros). All of these regressions control for the number of clean permissions and permissions that are needed to show ads. Heteroscedasticity-consistent standard errors are in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

number of permissions over the five-month period of observation. As in the business model, the price of an app is hardly ever adjusted. In both specifications of the cross section, we see an insignificant correlation between using privacy-sensitive permissions and prices. As discussed in Section 5, the cross section coefficient for permissions could be confounded by functionality and other factors. We can control for such unobserved heterogeneity in panel estimations, and indeed, we see a much stronger and weakly significant negative relationship in the panel. Although the scope of our panel estimation is limited by the small number of apps that can be included, this negative relationship is in line with the notion that apps might be trading privacy-sensitive permissions for a lower price (and vice versa).

Further supply-side results are shown in Online Appendix Table A4. In columns (1)–(4) we vary the specification in the supply-side regressions to highlight

further findings. Column (1) shows that apps are more likely to be free when the number of privacy-sensitive permissions increases. In column (2) we show that developers strategically use privacy-sensitive permissions in free apps only together with internet access, which highlights that data collection (via permissions) is more valuable for developers if they can easily transfer the data from the app (via internet access). Columns (3) and (4) use our category-specific privacy measure (see Online Appendix A.3 for more details on this measure). In columns (5)–(10) we contrast games and nongames, analyze moderating factors, and show the results for even more restrictively matched app pairs.

Taking all supply specifications together, we conclude that developers are trading access to privacy-sensitive information for money. Pay strategies are particularly associated with more privacy for the users, because paid apps request fewer sensitive permissions.

6.2. Demand-Side Analysis

6.2.1. Cross Section. Table 3 shows our main results for the relationship between app demand and the presence of privacy-sensitive permissions. We show (1) descriptive regressions for the full cross section (columns (1)–(3)), (2) fixed-effects panel regressions (columns (4) and (5)), and (3) a difference-in-differences-style (DiD) analysis (columns (6) and (7)). The demand measure (dependent variable) is the monthly change in the number of ratings (in logs). Column (1) shows the raw correlation between permissions and demand. Absent any control variables, the coefficients for both the privacy-sensitive permission dummy and the number of nonsensitive permissions have positive signs, which are presumably related to confounding factors. More permissions could be related to greater functionality, which leads to more demand. After we add control variables and controls for permissions that allow internet access and to show ads (column (2)), the positive coefficient of the dummy capturing privacy-sensitive permissions becomes negative (−6.5%) and

significant. By contrast, the coefficient of unproblematic permissions remains positive and significant throughout all cross section–based specifications. Also, the coefficient for ad-related permissions is positive, which points to the need for more rigorous estimation strategies, such as panel techniques. In column (3) we use the number of privacy-sensitive permissions as an alternative measure of potential intrusion. Also, with this measure, we find a significant and negative coefficient for privacy-sensitive permissions, whereas we find a positive one for unproblematic permissions.⁴² Both columns (2) and (3) document that privacy-sensitive permissions are associated with lower demand.

6.2.2. Panel Data Analysis. Columns (4)–(7) in Table 3 probe the robustness of our cross section results by accounting for the unobserved heterogeneity of apps. Columns (4) and (5) show the results of our fixed-effects panel regression analysis. We run a fixed-effects panel regression for apps that changed the number of privacy-sensitive permissions during our period of

Table 3. Main Demand-Side Results

	Cross section (log $\Delta Ratings$)			Panel (log $\Delta Ratings$)		Difference-in-differences	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
$D_{Privacy}$	0.012 (0.011)	−0.065*** (0.009)		−0.059*** (0.020)		−12.660* (6.634)	−14.988** (7.215)
$\#Privacy$			−0.021*** (0.003)		−0.014** (0.007)		
$\#CleanPerm$	0.111*** (0.004)	0.037*** (0.003)	0.041*** (0.003)	−0.010** (0.005)	−0.005 (0.005)	0.849 (0.643)	1.011* (0.605)
$D_{Internet}$		−0.200*** (0.009)	−0.211*** (0.009)	0.003 (0.054)	−0.014 (0.053)		
D_{Ads}		0.236*** (0.009)	0.230*** (0.009)	0.002 (0.033)	−0.014 (0.033)		
log $Price$		−0.071*** (0.001)	−0.071*** (0.001)	−0.044*** (0.011)	−0.043*** (0.011)		
Constant	−0.289*** (0.007)	−3.871*** (0.122)	−3.872*** (0.122)	1.198** (0.561)	1.201** (0.563)	4.262 (5.735)	2.428 (6.327)
Category	No	Yes	Yes	Yes	Yes	No	No
Controls	No	Yes	Yes	Yes	Yes	No	No
Month	No	No	No	Yes	Yes	No	No
Observations	177,193	177,193	177,193	33,095	33,095	192	162
No. of groups				6,619	6,619		
Mean of dependent variable	0.10	0.10	0.10	1.63	1.63	−0.24	−3.22
SD of dependent variable	1.55	1.55	1.55	2.23	2.23	31.81	29.27
Adjusted R^2	0.047	0.294	0.294	0.033	0.033	0.013	0.025

Notes. The table shows the relationship between the presence of privacy-sensitive permissions and app demand on three different data sets: columns (1)–(5) exploit our cross section and panel data; columns (6) and (7) show a difference-in-differences–style setup between Google's Play Store and the iOS App Store. In columns (1)–(5) the dependent variable is demand proxied by the log number of monthly new ratings of an app. Columns (1)–(3) contain cross section results. Column (1) shows the raw correlation between the use of privacy-sensitive permissions and demand. Column (2) adds in controls for the app's observed characteristics. Column (3) uses the number of privacy-sensitive permissions as privacy indicator. Columns (4) and (5) show panel fixed-effects regressions for those apps within our data set that varied their use of privacy-sensitive permissions at least once between April and September 2012. We show the results for the presence of privacy-sensitive permissions (column (4)) and their number (column (5)). In columns (6) and (7) the dependent variable is the difference between the app's download ranks on the iOS App Store vs the Google Play Store. In column (6) this difference is based on the average download ranks of seven countries, whereas in column (7) it is based only on U.S. ranks. All specifications control for the number of unproblematic permissions ($\#CleanPerm$). Heteroscedasticity-consistent standard errors are in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

observation. The results show that privacy-sensitive permissions are associated with lower demand, independent of whether we look at the presence of any (dummy) or the number of sensitive permissions. Similarly, the introduction of ads is no longer positive (unlike in the cross section, where ads are associated with greater demand, because of potential endogeneity).

6.2.3. Difference-in-Differences–Style Analysis. Our second approach to tackle unobserved heterogeneity is a DiD-style comparison of the same app's success in both the Google Play Store and in Apple's App Store. In columns (6) and (7) we compare the difference in apps' download rankings in Google's Play Store and Apple's App Store for apps that request sensitive permissions to the difference in the rankings for apps that do not request such permissions. Column (6) uses the aggregated rankings of seven countries, whereas column (7) uses only U.S. rankings. In both estimations, the value of the within-app rank difference between the two operating systems (iOS minus Android) is weakly significantly smaller for apps that require privacy-sensitive permissions, which indicates that apps with privacy-sensitive permissions perform relatively poorly on Android's platform.⁴³ Specifically, our results suggest a difference in the rank difference of the two groups of

apps equal to 12.6 ranks for the results based on the broader set of countries and a difference of 15 ranks for the U.S.-based results. Thus, both specifications confirm the cross section–based results, indicating a negative relationship between privacy-sensitive permissions and demand.

Above, we have presented our baseline result. Consumers have lower demand for apps that request privacy-sensitive permissions. We now move on to analyze moderators that amplify or reduce the size of this relationship (Table 4). Section 6.4 then provides an extensive series of robustness checks that aim to reduce the role of unobserved heterogeneity in the cross section. Also, we use alternative measures of demand and of privacy, apply alternative estimation specifications, and split our estimation sample into various subsamples.

6.3. Moderating Factors on the Demand Side

Previous literature (cf. Acquisti et al. 2016) has shown that the role of privacy concerns depends on an app's context. In this subsection we analyze how privacy concerns for mobile apps depend on their context. We take inspiration from a framework by Chellappa and Sin (2005) that highlights the trade-offs that consumers face when choosing how much personal information to share in exchange for better services. Chellappa et al. (2010)

Table 4. Moderating Factors of the Demand-Side Relationship

log Δ Ratings	Top apps		Privacy policy		Maturity level		Medical and health apps	
	No (1)	Yes (2)	No (3)	Yes (4)	High (5)	Low (6)	Yes (7)	No (8)
D_{Privacy}	−0.068*** (0.008)	0.008 (0.029)	−0.070*** (0.009)	0.038 (0.057)	−0.109*** (0.019)	−0.054*** (0.010)	−0.151*** (0.043)	−0.060*** (0.009)
#CleanPerm	0.029*** (0.002)	0.014*** (0.005)	0.033*** (0.003)	0.054*** (0.010)	0.028*** (0.008)	0.038*** (0.003)	0.029** (0.014)	0.036*** (0.003)
D_{Internet}	−0.122*** (0.008)	−0.320*** (0.040)	−0.192*** (0.009)	−0.254*** (0.063)	−0.108*** (0.019)	−0.211*** (0.010)	−0.106** (0.041)	−0.203*** (0.009)
D_{Ads}	0.188*** (0.008)	0.364*** (0.033)	0.229*** (0.009)	0.225*** (0.056)	0.292*** (0.025)	0.222*** (0.010)	0.180*** (0.042)	0.238*** (0.009)
log Price	−0.048*** (0.001)	0.003 (0.007)	−0.070*** (0.001)	−0.114*** (0.005)	−0.050*** (0.002)	−0.076*** (0.001)	−0.056*** (0.003)	−0.072*** (0.001)
Constant	−3.470*** (0.106)	−4.694*** (0.499)	−3.753*** (0.122)	−4.497*** (0.976)	−3.467*** (0.247)	−3.980*** (0.140)	−3.811*** (0.669)	−3.865*** (0.125)
Category	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	164,157	13,036	170,658	6,535	27,843	149,350	6,218	170,975
Mean of dependent variable	−0.14	3.16	0.06	1.11	−0.18	0.15	−0.08	0.11
SD of dependent variable	1.27	1.54	1.52	2.02	1.38	1.58	1.38	1.56
Adjusted R^2	0.221	0.293	0.285	0.371	0.305	0.291	0.270	0.295

Notes. The table shows the relationship between the presence of privacy-sensitive permissions and app demand for subsamples of our data. App demand is measured by the log number of monthly new ratings of an app. Columns (1) and (2) split the sample into apps that have a high and a low stock of installations (more or less than 75,000 installations), respectively. Columns (3) and (4) split the sample into apps with and without a privacy policy, respectively. Columns (5) and (6) split them into apps that require a high (column (5)) or low (column (6)) maturity of the user (apps are defined as appropriate for low maturity if they are classified as being recommended for “everyone” or for “low maturity” users). Columns (7) and (8) split the sample into medical and health-related apps (column (7)) and into other apps (column (8)). All specifications control for the number of unproblematic permissions (#CleanPerm). Heteroscedasticity-consistent standard errors are in parentheses.

*** $p < 0.01$; ** $p < 0.05$; * $p < 0.1$.

embed these trade-offs in a formal mechanism design problem that captures three key moderators of aggregate privacy concerns that arise in our empirical context: (i) the composition of consumers, (ii) trust-inspiring measures by firms, and (iii) the nature and amount of the requested private information.⁴⁴ Three predictions can be derived from this model (and the broader privacy literature in general). First, if developers can implement trust-inspiring measures, privacy concerns are reduced; that is, users are more willing to adopt products that request sensitive information. Second, we expect differences in the willingness to share sensitive information among subgroups of the total population of users—specifically, we expect mature users to be more aware or cautious than younger users in adopting privacy-sensitive apps. Third, and finally, users might more hesitantly adopt apps with the ability to collect personal information if they operate in sensitive contexts, in which the shared information might be abused or used to discriminate against the consumer. Examples of such contexts are health-related or financial apps.

In Table 4 we highlight examples in which trust (columns (1)–(4)), the targeted user group (columns (5) and (6)), and the sensitivity of the shared information (columns (7) and (8)) indeed moderate the relationship between privacy sensitive permissions and app demand: for each of these factors, we separately estimate apps of a specific type (e.g., apps that have a privacy policy) and apps that do not satisfy the criterion. The coefficient of interest is D_{Privacy} , and specifically, we compare the coefficient for two groups of apps and analyze whether it is equal for both groups or not. This analysis serves a double function. First, if privacy concerns drive our results, the relationship of interest should vary by context as theory predicts. Second, we can generate additional insights into how and in which contexts privacy concerns matter in the market for mobile apps.

In columns (1)–(4) of Table 4, we analyze the role of apps' past success and trust-enhancing measures such as privacy policies. Columns (1) and (2) study how the effects vary for top apps with a very large user base (i.e., apps with more than 75,000 installations). Among apps with a very large user base, we observe no negative correlation between the presence of privacy-sensitive permissions and demand, whereas for less successful apps, the negative effect prevails. Columns (3) and (4) estimate how the relationship of interest differs for apps that are transparent about their privacy policy. Apps that publish a directly accessible privacy policy in the Google Play Store would be expected to inspire additional trust from app users. Indeed, among apps with an easily accessible privacy policy, sensitive permissions have no statistically significant effect.⁴⁵ Columns (5) and (6) in Table 4 analyze the role of

privacy concerns for different user groups. Columns (5) and (6) look at how the coefficient estimates differ for apps that are not suitable for children or young adults. Privacy might be a greater concern in such apps, because they target a different group of users, who are, on average, more privacy sensitive.⁴⁶ The significantly more negative coefficient for D_{Privacy} in column (5) highlights that mature users (or no indication) avoid privacy-sensitive permissions more actively, whereas apps that are suitable for children are less punished for using privacy-sensitive permissions.⁴⁷ In columns (7) and (8) we separately analyze the most sensitive category of apps, medical and health-related apps, because we would expect the relevance of privacy concerns to differ for this category.⁴⁸ Indeed, we see that users of medical and health-related apps seem to avoid privacy-sensitive permissions more than users of other genres.

This subsection highlights that reputation, the app's target group, and the sensitivity of the app's context moderate the negative relationship between using privacy-sensitive permissions and app demand. We found the strongest positive moderating effect for well-known apps and for apps that adopt a privacy policy. The strongest negative moderation was found for health-related apps. In Appendix Tables A7 and B5, we show the robustness of these results to considering alternative moderating factors or different specifications.

6.4. Robustness

We begin this section by discussing the results of our alternative supply-side estimations (Online Appendix Table A4). Specifically, we show that the main results do not change if we use more rigorous matching or subsamples and that the moderating factors from the demand-side analysis are also present on the supply side. We then move on to the robustness of our demand-side results. We show additional panel specifications (Online Appendix Table A5) and more DiD results (Online Appendix Table A6). We also analyze alternative demand measures (e.g., installation-based measures; Online Appendix Table A8), different ways of measuring privacy-sensitive permissions (Online Appendix Tables A9 and A14), employing alternative estimation strategies (Online Appendix Table A10), and considering/omitting subsegments of the market (Online Appendix Tables A7 and A11).

6.4.1. Supply Side.

The Supply Side Is Robust to Using Alternative Definitions of Privacy and Subsamples. In Online Appendix Table A4 we test the robustness of our supply-side results that analyze the business model choice; that is, the dependent variable is again the dummy equal to 1 if an app is a paid version. In columns (1)–(6) we varied the specification in the supply-side regressions

to verify both that our results do not depend on our privacy variable and that they hold within subsamples of our data such as games and normal apps. The main cross section results do not depend on the choice of the specific privacy measure. For increasing numbers of privacy-sensitive permissions, apps are more likely to be free (column (1)). Column (2) shows that developers use privacy-sensitive permissions in free apps only together with internet access, highlighting that data collection (via permissions) requires internet access to transfer the data from the app. In columns (3) and (4) we use our category-specific privacy measure and show that our results are robust to this context-specific definition of privacy intrusiveness (see Online Appendix A.3 for more details on this measure). Columns (5) and (6) show that our baseline finding holds for both nongame apps (column (5)) and games (column (6)). For both groups, we find a negative significant effect.

Supply-Side Results Are Robust to Including Moderating Factors and to Using More Rigorously Matched App Pairs. In columns (7)–(10) in Online Appendix Table A4, we analyze the role of moderating factors on the supply side and restrict the pairs data set even further to rule out unobserved heterogeneity as an explanation of our results. Columns (7) and (8) show that the role of reputation as a moderating factor is just as influential for the supply side as for the demand side. In column (7) we separately analyze apps with a large user base (more than 10,000 installations), and in column (8) we analyze apps that are associated with a popular website (low traffic rank on [Alexa.com](https://www.alexa.com)). Such apps are generally less likely to be paid versions, but if they are paid, they are more likely to require privacy-sensitive permissions. Columns (9) and (10) analyze the robustness of our supply-side results from the app pair data by applying more restrictive matching for the pair data: here, we consider only pairs with no difference in description and code length, which was verified by human coders. These results show again that privacy-sensitive permissions are more likely in free apps, independent of how restrictive we are in our matching of the app pairs.

Taken together, these results confirm our baseline findings for the supply side in Table 2.

6.4.2. Demand Side.

Panel Estimations Are Robust to Using Alternative Samples and Specifications. Next, we run alternative panel fixed-effects regressions, which use an alternative set of controls, employ alternative data sets, and vary the explanatory variable of interest. These checks ensure that our demand-side results do not critically depend on our choice of controls, the sampling, or the variable of interest. In Online Appendix Table A5

we show these additional panel fixed-effects regressions. The estimated coefficient of interest remains essentially unchanged for the specification in column (1), where we include only a reduced set of controls (we include only alternative sets of permissions) or when the data are limited to apps that introduced permissions, but no major update to functionality, during our period of observation (column (3)). Moreover, we see similar results when analyzing the number of sensitive permissions, rather than an indicator of their presence (columns (4)–(6)). Finally, the panel results are also robust to using a category-specific definition of privacy-sensitive permissions, as shown in Online Appendix Table A14. Altogether, the main results are confirmed, and the estimated effect of introducing permissions is even a bit larger when we restrict the sample to apps without any improvements in functionality.

DiD-Style Comparison Is Robust to Using Alternative Measures of the Ranking Difference. In Online Appendix Table A6 we test the robustness of our DiD-style setup to alternative methods of computing the difference in iOS and Android rankings as well as to splitting the sample into game and nongame apps. As before, the dependent variable captures differences in the download rankings of an app on the two platforms (iOS App Store rankings minus Google Play Store rankings). Columns (1), (2), (5), and (6) compare the apps' average rankings in the seven countries for which we collected rankings, whereas columns (3) and (4) use only U.S. ranks. The dependent variable in columns (1) and (3) is the difference between the simple ranking averages, whereas in columns (2) and (4) we use the difference between the newly created in-sample ranks within the OS (as in the baseline demand table), which is based on the average download rankings we observed on [AppAnnie.com](https://www.appannie.com). Columns (5) and (6) also use the difference between the in-sample rankings based on the average download ranks but contrast games (column (6)) with other apps (column (5)). All specifications except specification (6), which covers only games, show a significant negative effect for privacy-sensitive permissions.⁴⁹ This corroborates that apps with privacy-sensitive permissions are on average more successful in the Apple iOS than in the Android OS, where these permissions are visible to the user before the app is installed. Thus, the results reaffirm our baseline demand results despite using a completely different sample of apps and a completely different identification approach.

Demand-Side Results Are Robust to Using Alternative Demand and App Popularity Measures. In Online Appendix Table A8 we verify that our main findings for the demand side do not depend on our preferred demand measure. Specifically, we estimate our main

specification with 10 alternative demand measures based on ratings (columns (1) and (2)), the direct measure of installations (columns (3)–(5)), and 3 measures of predicted monthly new installations (columns (6)–(8)). Our main demand-side results remain the same, independent of whether we use measures based on installations or measures based on ratings or of how we vary the time window of ratings or installation growth (columns (1)–(5)).

Similarly, when we use predicted changes in the number of installations (columns (6)–(8)), our results are also confirmed. In columns (6)–(8) we apply three measures of predicted installation numbers. For each of the measures, we exploit the cross section information on changes in ratings to predict changes in installation numbers. In column (6) we predict monthly installation changes in April 2012 based on the observed change in the number of ratings. In column (7) we predict installation changes between April and September 2012 based on the observed change in ratings during this period. For both specifications, we use a parametric log-log-specification to predict the change in the number of installations (cf. columns (2) and (4) of Online Appendix Table A12). In column (8) of Online Appendix Table A8, we use a measure of nonparametrically predicted monthly installations in April 2012.

Finally, in columns (9) and (10), we study how the existence of privacy-sensitive permissions affects apps' popularity as measured by their average ratings. In column (9) the dependent variable is the average rating in April 2012 (which is between one and five stars), whereas in column (10) it is the average rating between April and September 2012. Both specifications indicate that apps receive worse ratings if they use privacy-sensitive permissions.

Demand-Side Results Are Robust to Using Alternative Privacy Measures. We verify that our results do not depend on our workhorse definition of what is considered "privacy sensitive" by estimating our main specification with multiple alternative definitions. Specifically, in Online Appendix Table A9 we show our main regression for eight alternative privacy measures. These measures include Google's own classification, the definition by Sarma et al. (2012), a category-specific measure, and one derived via classifiers from Amazon Mechanical Turk. The results show a negative relationship between privacy-sensitive permissions and demand across six of the eight measures that we consider, and in column (7) only one out of four permission groups has a positive coefficient.⁵⁰ The only measure with a positive relationship is the *privacy grade* provided by Lin et al. (2014), which grades apps' privacy intrusiveness in 2014 for a subsample of our

data. The positive coefficient indicates that demand is higher for apps with a bad privacy grade. This finding was unexpected but could be explained by the fact that the privacy grades are based on analyzing app's software libraries, which is possible only if apps use standard software libraries. Using such libraries might require a certain level of experience and professionalism and thus might pick up unobserved developer quality.

As in our baseline results, D_{Ads} and the number of unproblematic permissions are positively associated with demand in all specifications, which indicates that unproblematic permissions, unlike privacy-sensitive permissions, do not lead to lower demand but might capture functionality and therefore might have a positive coefficient.⁵¹ We also would like to highlight the results in column (7), where we distinguish between permissions by type. Two of four permission groups have a negative, significant coefficient (D_{Location} , D_{Profile}), one coefficient ($D_{\text{Communication}}$) is insignificant, and one (D_{ID}) seems to be positively associated to demand. Even though they are based on an analysis of the cross section, these findings might indicate that users have different degrees of willingness to share certain types of information and warrant further research using individual user-level data. Moreover, we analyze category-specific privacy-sensitive permissions in additional depth. The "category-specific" definition takes into account that privacy is context dependent and considers that apps in a certain category might need access to certain data to perform their services.⁵² In Online Appendix A.3 we discuss the measure, and especially in Online Appendix Table A14, we show the main demand-side results for the cross section and panel also hold if we use the category-specific definition.

Demand-Side Results Are Robust to Accounting for Censoring, Selection, Network Effects, and Endogeneity. We also run the main regressions with alternative estimation methods to verify that our demand-side findings are not merely due to our OLS specification. In Online Appendix Table A10 we use Tobit estimation to account for the possibility of censoring in the dependent variable (columns (1) and (2)). In addition, in columns (3) and (4) we provide results from Heckman selection models, where we model apps' survival between April and September 2012 (column (3)) as well as apps' survival between 2012 and 2014 (column (4)). We also control for potential network effects (column (5)) and introduce instruments for potential endogeneity in prices and developers' use of permissions (columns (6)–(9)). None of these alternative specifications drastically changes our baseline findings.

Sample Splits and Success. In Online Appendix Table A7, we show that alternative moderating factors highlight similar patterns as documented in Table 4. The table analyzes the moderating role of the following factors: (1) top developer-status (columns (1) and (2)), (2) apps connected or not connected to a well-known website (using Alexa ranks) (columns (3) and (4)), (3) games versus nongames (columns (5) and (6)), and (4) pricing strategy (columns (7) and (8)). Also, for these moderating factors, the strength of the effect varies along similar lines as in our main results.

Next, we test whether our main result is robust to excluding the most and the least successful apps.⁵³ This robustness check allows us to exclude the possibility that only a few apps drive our results and provides additional insights into the anatomy of our main demand-side results. Online Appendix Table A11 shows that excluding the most successful apps (5% of apps with the highest demand; columns (1) and (2)) and excluding both the most and least successful apps (those without any new rating in April 2012; columns (5) and (6)) does not change our findings. We found an insignificant permission coefficient only by disregarding the least successful apps (columns (3) and (4)). However, this finding is not robust to including a dummy that equals 1 if an app already has a user base of more than 10,000 installations as well as an interaction term with privacy-sensitive permissions. Doing so, we also find a negative significant effect for privacy-sensitive permissions for that sample. The dummy and the interaction term together capture the influence of an app's reputation (as in the analysis of moderating factors). This finding corroborates the role of reputation as a moderating factor and suggests that large and established apps may have easier access to sensitive user data than other players. Unknown apps face a penalty for requesting privacy-sensitive information.⁵⁴

6.5. Implied Valuations

We now turn to discussing the value of privacy-sensitive permissions, as implied by our demand and supply estimates. We do so by comparing the estimated demand coefficient for privacy-sensitive permissions with the own-price elasticity of apps. Both higher prices and the use of sensitive permissions are associated with lower demand. From the cross section results, the coefficient for apps that use sensitive permissions is about 0.9 times the size of that corresponding to a 1% increase in price. The panel estimates indicate that ratio is 1.5, suggesting that the penalty for using sensitive permissions is equal to the penalty corresponding to a 1.5% higher price. The average price of paid apps of 2 euros implies a lower bound "willingness to avoid" of 2 to 3 euro cents. We reiterate, however, that our work suffers from limitations, and we consider our coefficients to be lower-bound

estimates of the effect of introducing such permissions, because of the endogeneity concerns that result from imprecise demand measures and unobserved app quality. Hence, the implied valuations should be used carefully, and any policy implications require validation with individual-level or experimental data.

On the developer side, a first meaningful way of quantifying the valuation of data can be based on columns (1) and (2) in Table 2, where we observe that a privacy-sensitive permission is associated with a 15% higher probability that an app is free. Multiplying this by the average price of a paid app (2 euros) gives an estimate of 30 euro cents. We find a similar value if we use the results from the price choice model (Table 2, column (9)). The coefficient of privacy-sensitive permissions (-0.118) indicates that developers are willing to reduce the app price by about 12% if the app has a privacy-sensitive permission. This implies a price reduction of 24 euro cents for the average paid app. Again, these estimates suffer from severe limitations, and they are likely confounded by other sources of income that the free apps can generate (building a user base, etc.). We leave it to future research to evaluate the developers' valuations more precisely.

7. Discussion

We document that private information plays a critical role in the market for mobile apps, because it resembles a second "means of payment." On the supply side, app developers ask for more privacy-sensitive permissions in exchange for a free app, or they offer a less privacy-sensitive app for a higher price. On the demand side, we observe fewer installations for apps that request more privacy-sensitive permissions. Together, these results highlight a money-for-privacy trade-off in the market for mobile apps.

We use various subsets of data to obtain our results: a full cross section, panel data, carefully selected app siblings, and a DiD-style strategy that compares the same app on the two biggest app platforms. We can show the relationship for both the supply and demand for apps, and our extensive set of robustness checks highlights the robustness of our findings. Our conclusions persist across (1) different ways of quantifying intrusiveness, (2) several alternative classifications to measure the sensitivity of permissions (Google's own classification, the definition by Sarma et al. 2012, a category-specific measure, and one derived via classifiers from Amazon Mechanical Turk), (3) several conceptualizations of demand (ratings, installations, first-differences, etc.), and (4) several alternative estimation approaches.

Two patterns emerge from our analysis. First, we find economically significant demand-side effects.⁵⁵ To be precise, our (lower-bound) estimate of consumers' willingness to avoid sharing personal information is

between 2 and 3 euro cents. These estimates are between the extremes of the privacy valuations found in experimental work and other contexts (Grossklags and Acquisti 2007, Beresford et al. 2012, Carrascal et al. 2013). However, our estimates are considerably smaller than previous estimates for apps based on surveys (Savage and Waldman 2015), even though the Android OS makes the information very explicit. The potential explanations include the following: (1) Users might simply show behavioral biases and overstate their preference for privacy in surveys or experiments. (2) We provide only lower-bound estimates of the users' true reaction to privacy sensitive permissions because of data limitations in our demand measure and the challenge of fully isolating the effect of privacy sensitivity from that of functionality. (3) Having too much unstructured information about permissions might be like having no information at all. Whereas the first explanation stresses the importance of using observational data for appropriately estimating consumers' valuations, the third hypothesis could align our small coefficients with the findings of Savage and Waldman (2015), who found that consumers valued an app's access to private information much more negatively, but only *after* receiving information about the permissions' implications. To test whether information overflow is the underlying reason, simple pieces of summary information should suffice to reduce the gap between earlier findings and ours.⁵⁶

The second key pattern duplications of analysis emerges from our analysis of moderating factors. We confirm users' generally negative attitude toward privacy-sensitive permissions but also show that this finding does not always hold. The strength of the relationship depends on the context as well as app and user characteristics, which confirms previous research (Chellappa and Sin 2005; Goldfarb and Tucker 2012; Acquisti et al. 2013, 2016). For example, the negative effect of permissions is weaker if privacy-sensitive permissions are common in an app category or if it is likely that a consumer is relatively young. However, the negative relationship between demand and permissions can also become stronger. If an app is in more sensitive categories (such as health-related apps), intrusive permissions are associated with even lower demand. This finding suggests that users are aware of the value of specific types of personal information and share it more cautiously. Trust appears to be the most prominent factor, as is documented by the insignificant effect of sensitive permissions on apps that adopt a privacy policy and on widely adopted apps that already have a large user base. Supplementary results show similar patterns for apps by top developers or apps that can be associated to widely used websites (via [Alexa.com](https://www.alexa.com)). Consumers who care but do not fully understand all permissions might create this pattern,

because they might rationally prefer apps that are well known to those that are unknown. On the positive side, such behavior allows established firms to improve their products based on consumer data, which can increase quality. On the negative side, users' reliance on outside reputation might create a barrier to entry: well-established apps can gather user information more easily than newcomers, reinforcing the arguments in Campbell et al. (2015) and potentially reducing innovation in the app market.

Our work suffers from several limitations, which offer avenues for further research. Although we do our best to scrutinize the robustness of our main results, we do not have ideal data. We do not have access to the exact numbers of installations and even more precise information on an app's characteristics, quality, or use of ads and have to approximate these variables. Approximation carries the risk of picking up confounding influences from ads or unmeasurable aspects of quality. We address these concerns by intensely testing the robustness of our findings, and we further validate our main findings on more rigorously matched subsamples. However, these strategies involve their own compromises, such as smaller data sets, the risk of attenuation bias, and lower significance levels. Future work could improve our research by using the exact installations numbers or daily active users, which is developers' most preferred measure of usage. Moreover, we cannot completely eliminate two problems: (1) Time-variant unobserved heterogeneity cannot be controlled by a panel fixed effect.⁵⁷ (2) When conditioning on apps without functionality improvement, we fail to observe developers who introduce sensitive and unnecessary permissions alongside a new feature. This caveat reduces the power of our specification. Similarly, the analyzed moderating factors do not change over time, which disallows a panel analysis. Hence, these additional results are based on conditional correlations in the cross section. For example, the weaker negative relationship between permissions and demand for younger consumers should be validated with user-level data. Such information is needed to decide whether young users install apps with too little reflection or whether older users are too cautious. Hence, great care is warranted before any policy changes can be implemented in this potentially sensitive two-sided market, and further research must carefully evaluate the effects of policy changes that aim to highlight the potential intrusiveness of an app. Finally, further research should analyze developers' dynamic strategies to gain access to data in the market for mobile apps.

In sum, we see our findings as a first step in understanding the role of privacy in app markets. Any policy implications that we suggest should be validated with individual-level or experimental data. We believe that such a careful evaluation would be a fruitful avenue

for further research. Hence, significant unleveraged potentials might exist to further improve the performance of the market, especially in sensitive categories.

8. Conclusion

The so-called American perspective on privacy views it as a property right (Camp 1999). This perspective is interesting to managers, as it suggests that private information can be negotiated between firms and consumers. Our paper speaks to this view and documents trade-offs with respect to privacy-sensitive information for both firms and consumers in the market for mobile apps. The app market is economically relevant and offers previously unknown potential for collecting information about consumers. A money-for-privacy trade-off in this market matters, because it suggests a cost for greater privacy protection, independent of whether such protection is desired.

We base our evidence on large observational data that reflect revealed preferences. We analyzed information on nearly all the apps in Google's Android Market in 2012 (approximately 300,000 apps). We augmented the data with additional information from [Alexa.com](#), [PrivacyGrade.org](#), and [AppAnnie.com](#). Using these data, we study the role of privacy in both the supply and demand for mobile apps. Specifically, we study (1) whether developers use app permissions to collect private information about users (e.g., information about their communication behavior, location, and profile) and (2) how consumers' installation behavior is related to privacy-sensitive permissions.

Using several data sets, with alternative measures of intrusiveness and multiple specifications, we document that private information plays a prominent role on both sides of this market: app developers offer either a lower price for a more privacy-sensitive app or a higher price for more privacy. Consumers choose between these two options (privacy and money), and we indeed find a small but robust and economically significant negative relationship between permissions and demand.

In addition, the sensitivity of consumers toward apps' permissions was found to depend on the app's context: demand is lower for apps with content that is rated as appropriate for older users, or in more sensitive categories, such as health-related apps. The negative correlation between demand and permissions disappears for known apps and apps that adopt a privacy policy. This differential behavior is in line with consumers who find it difficult to evaluate whether they are installing a privacy-endangering product and react by favoring brands and apps that have been widely adopted. Yet we also found more cautious behavior in sensitive contexts, such as health, which suggests that consumers are considering costs and benefits when sharing their information.

Our results suggest that the app market is a favorable environment for established app developers who seek to collect information about their consumers. The flip-side interpretation of this result is that the resulting lack of trust of unknown app developers might constitute a significant barrier to entry. Such a barrier could point to suboptimal innovative performance of the market for mobile apps especially in "sensitive" categories, such as health-related apps. If consumers lack sufficient knowledge to rely on their judgment regarding privacy-sensitive permissions, a simple certificate or label by a third party (e.g., a traffic light symbol) might help to reduce this entry barrier.

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Endnotes

¹ According to the Pew Research Center, 68% of adults believe current laws to protect individuals' online privacy are insufficient (Rainie et al. 2013).

² We refer to real historical experience (e.g., in Europe or Asia) where too much data in the wrong hands contributed to totalitarianism, persecution, mass murder, and war.

³ We use "second means of payment" metaphorically, because both parties agree to exchange the developer's service for the user's data, and then developers exchange user information for money or services of third parties (comparable to a complementary medium of exchange, a tradable asset, or a valuable resource). We do not suggest that private data satisfy the definition of a full-fledged and widely accepted currency, with established exchange rates.

⁴ For several robustness checks, we further augmented our database with information from [Alexa.com](http://www.Alexa.com) and data collected using Amazon's Mechanical Turk.

⁵ Acquisti et al. (2016) and Tucker (2012) provide surveys. Taken together, these models find that reduced privacy is disadvantageous for consumers. Firms can use customer information, such as purchase history, to charge personalized prices in electronic retailing settings (Taylor 2004, Acquisti and Varian 2005, Conitzer et al. 2012). However, increasing the cost of anonymity can benefit consumers up to a point, after which the effect is reversed (Conitzer et al. 2012). Spiegel (2013) shows in the context of software production that free software in a bundle with (targeted) ads could improve welfare if the cost of producing software is low. Reduced privacy enables the provision of valuable services “for free” and can create benefits for users.

⁶ Chellappa et al. (2010) set up a mechanism design problem in which a monopolist faces consumers who have heterogeneous privacy concerns and decides on an optimal menu of personalization with possible subsidization. Casadesus-Masanell and Hervas-Drane (2015) study the money-for-privacy trade-off in a model in which suppliers compete on both dimensions (price and privacy) of a two-sided market, as pioneered by Armstrong (2006) and Rochet and Tirole (2003).

⁷ For example, by 2008, the self-reported valuations for protection against improper access and secondary use were found to range from USD 30 to USD 50 (Hann et al. 2007, 2008). High exposure of personal information through a smartphone can lead to identity theft, and users' behavior can even affect the privacy of uninvolved friends, or the information security of their company by increasing the risk of successful “choice and chance” attacks (Ransbotham and Mitra 2009, Pu and Grossklags 2016).

⁸ Specifically, restricting the use of private data in advertising can reduce the success of targeted advertising and the revenue generated. More generally, websites differ considerably in how intensely they collect user data, and privacy policies can affect user behavior (Johnson 2013a, Preibusch and Bonneau 2013).

⁹ Pioneering papers focused on identifying key drivers of app demand and estimating their importance. Other recurring themes are the relationship between innovation and quality in app markets, developers' strategic behavior, or their strategies for establishing a new app in the market (e.g., paying users to download the app onto their devices in order to obtain a position on the list of the most downloaded apps). Structural models analyzed demand and app developers' resulting platform choice (Android or iOS).

¹⁰ They use a valuation scheme based on providing information about privacy relevant aspects of the data and ask users their willingness to accept. They also cannot say much about the developer dimension.

¹¹ Specifically, we show that consumer demand responds very little to potentially privacy-endangering permissions, which is in line with considerably lower consumer valuation for privacy and implies our conclusion.

¹² More technical studies have been done on the precise meaning of certain permissions and what they imply for the privacy of the device's owner (see, e.g., Chia et al. 2012 and Sarma et al. 2012), or they investigate the potential intrusiveness of apps (e.g., Chia et al. 2012 and Fahl et al. 2012). Other studies investigate how platform architecture affects smartphone users' willingness to pay premiums to limit exposure of their personal information (Egelman et al. 2013). Sutanto et al. (2013) suggest storing user information on the device to overcome the personalization versus privacy paradox and document a positive effect on demand. Two studies by Lin et al. (2012, 2014) analyze more than one million apps' privacy-related behaviors. They summarize these behaviors using a grade scale, ranging from A+ (most privacy sensitive) to D (least privacy sensitive; see <http://privacygrade.org/faq>, accessed November 5, 2016).

¹³ Even though our lower-bound estimates suggest lower valuations than previous studies.

¹⁴ As of March 2017, the count stands at 148 permissions (although the precise contents of some permissions changed; see <http://developer.android.com/reference/android/Manifest.permission.html>, accessed October 8, 2018).

¹⁵ Although requested permissions were displayed slightly differently on smartphones, the text of the permissions' descriptions is identical in both versions (mobile and browser) of the Play Store. The layout is almost identical.

¹⁶ Since 2016, Google allows users to withdraw individual permissions from an app after installation. However, this only affects recent versions of the Android OS (starting with version 6.0).

¹⁷ Android's app developers receive 70% of the app price, and 30% goes to distribution partners and operating fees (see <https://support.google.com/googleplay/android-developer/answer/112622?hl=en>, accessed October 8, 2018).

¹⁸ In 2011, Google added in-app billing to Android Market, allowing apps to sell in-app products (see <http://android-developers.blogspot.de/2011/04/new-carrier-billing-options-on-android.html>, accessed October 8, 2018).

¹⁹ The pieces of information most common transmitted about the user are identifying information (name, email address, phone ID, gender, age, or birthdate), location data, contacts, and usage data. Sometimes “data input” (e.g., search terms) is transmitted as well. This information is more sensitive in some contexts, such as health, than in others (Schmidt et al. 2012, Appthority 2014, Zang et al. 2015). Most of these studies are based on very small samples. For example, Seneviratne et al. (2015) studied the apps installed on the phones of 338 users and identified 124 different trackers in 509 unique apps in Australia, Brazil, Germany, and the United States. Trackers were categorized as advertising (e.g., Google Ads, Millennial Media, InMobi, MoPub), analytics (e.g., Flurry, Google Analytics, ComScore, Health Insights, Amazon Insights, Localytics, Kontagent, Apsalar), and utilities (e.g., Crashlytics, BugSense). Moreover, they found that 50% of these users were exposed to more than 25 trackers.

²⁰ Successful apps with sufficient traffic are able to sell their advertising space directly on the market. However, doing so is costly for less successful apps, which can achieve better targeting by sharing their data with a third-party broker. Such data brokers can provide advertisers with access to users from a bigger pool of multiple apps (and developers to more advertisers).

²¹ Peer-reviewed estimates are not available. Existing estimates range from several hundred million to several billion U.S. dollars (Organisation for Economic Co-operation and Development 2013, based on Beales 2010).

²² To use ads (even without targeting), developers need access to a phone's network state to simplify communication between the ad library and the ad server. For targeting consumers, developers frequently access a phone's fine location and permission, which allows for identifying the app's user (READ_PHONE_STATE; see, e.g., Book et al. 2013).

²³ The discrete measure introduces attenuation bias through noise in the dependent variable.

²⁴ Purchased ratings or installations could be problematic for our empirical strategy for two reasons: (1) widespread fraudulent reviews or installations could lead to considerable noise in our dependent variable, which might result in attenuation bias, and small estimates; and (2) if apps consistently differ in their strategies to encourage ratings and in their data collection behavior, our results might be biased. Specifically, if apps that use more permissions more actively recruit raters, this would bias our coefficients toward zero. To address the first concern, we compared

installations and ratings and found little evidence for fake ratings as a widespread phenomenon. “Hiring” ratings is most effective for young (Muchnik et al. 2013) and sometimes intermediate apps, which begin to have imitators (copycats). However, it is less likely to have an effect on monthly changes in later periods. We address the second issue, which is of critical importance, in our panel fixed-effects specifications, but it may affect the cross section estimation with attenuation bias. We also do not see many such sudden peaks in ratings over time.

²⁵ In Online Appendix A.2 we discuss the close relationship between our main demand measure and our measures based on app rankings. However, app rankings are available only for the most successful apps, which limits the scope of this comparison with a small and arguably selected set of apps.

²⁶ D_{Ads} equals 1 if an app uses the ACCESS_NETWORK_STATE permission (see Online Table A1). About 80% of ad libraries in 2013 used this permission, which simplifies communication between ad library and the ad server (see, e.g., Book et al. 2013). For the provision of targeted advertisement, especially READ_PHONE_STATE (which allows identifying app users) and ACCESS_FINE_LOCATION seem to be of high relevance.

²⁷ Introducing sensitive permissions in the free but not the paid version is the most explicit instance of a money-for-privacy trade-off. A typical, although not ideal, pair consists of a free “lite” version and the full (paid) version of the same app, in which the free version has more permissions but less functionality. In our research, we can use pairs to find unnecessary permissions, because the paid version is the proof that the same app can run without that permission. This inference is valid even if the paid version of the app offers more features than the free version, if only the paid app does not offer less functionality.

²⁸ By “functionality,” we mean the utility-generating service to the user. Ads, for example, might imply code and functionality, but their purpose is generating (data-driven) revenue for the developer (not increasing user satisfaction).

²⁹ According to our procedure, a pair’s free app is of the same code size or a little larger (on average, 300 KB larger). In the full cross section, the free apps are smaller (on average, 3 MB smaller), so we attribute the difference in the matched pairs to the developer’s monetization strategy. A pair’s length of description differs by less than 30%.

³⁰ In addition to overall rankings, we include category-specific rankings (e.g., for games, weather, or education apps).

³¹ Specifically, we compute average rankings by app and OS based on time- and country-specific ranks (up to seven countries and two points in time per OS—i.e., up to a total of 14 ranks). Using these average values, we create a new in-sample ranking that ranks apps by the order of their average rank values and ranges from 1 to 192 for each OS (because we have 192 apps available). In the estimation, we use the difference between the standardized rank we generated for the iOS version and the one from the Android version (i.e., iOS rank minus Android rank).

³² The discrepancy between our sample and the full population of approximately 300,000 apps is mainly due to excluding apps that were unavailable in some of the monthly waves in our panel analysis, lacked key variables, had zero installations or ratings, or are outliers with respect to our main demand measure.

³³ Because a free sibling is available, it is hard to imagine that users would be willing to pay a positive amount for less functionality, such that if two siblings have any difference in functionality, we assume that the paid app sibling provides more functionality.

³⁴ Apps with privacy-sensitive permissions are $\beta \times 100\%$ less likely to be paid.

³⁵ For any remaining differences in functionality, our results would continue to be biased toward zero if greater functionality is in the paid

version and is positively correlated with price or more (privacy-sensitive) permissions.

³⁶ A 1% price change leads to a $\gamma\%$ change in expected demand.

³⁷ Given product quality, neither code length nor the number of products by a developer should generate utility for the user but reflect the production cost and developer experience as well as potential for code sharing. As an alternative, “Hausman instruments” could be used in the panel, but the variation in prices is too small. Furthermore, note that free apps have a constant (zero) price, such that no instrumentation is required in their case.

³⁸ However, if permissions are introduced at a later point, even a panel approach could fail. In our favor, the endogeneity problem is somewhat mediated by the fact that functionality and offering pairs of apps are strategic choices. Developers must make these choices and introduce the implied permissions before demand can be observed.

³⁹ As described in Section 4, we provide two versions of this measure, which are based on average OS-specific download ranks (1) according to the average rank in seven countries and (2) exclusively according to U.S. ranks.

⁴⁰ We thank an anonymous referee for suggesting this specification.

⁴¹ We find a coefficient equal to 0.024, which suggests that each additional permission increases the probability that the app is free by 2.4%. When using the dummy D_{Privacy} and the number of permissions $\#Privacy$ together, we find the dummy to be insignificant and the coefficient of $\#Privacy$ still equals 0.024. We further investigate the marginal effect in Online Appendix Table A4.

⁴² The coefficient for $\#Privacy$ of 0.021 suggests a 2.1% lower demand for each additional permission. However, when estimating the dummy D_{Privacy} and the number of permissions $\#Privacy$ jointly, we find a significant coefficient of -0.049 for D_{Privacy} , and the coefficient of $\#Privacy$ is significant at -0.011 , which suggests a nonlinear effect, which we further investigate in column (1) of Online Appendix Table A9.

⁴³ Already, the raw mean values of the OS-specific rankings confirm this finding: the mean rank of apps with (respectively, without) privacy-sensitive permissions in the iOS is 50.2 (respectively, 58.5), whereas in the Android OS these mean values equal 52.4 (with such permissions) and 52 (without such permissions). Thus, in the iOS, apps with privacy-sensitive permissions have a lower mean rank; that is, they are more successful than apps without such permissions. By contrast, in the Android OS, if one group has a higher mean rank, it is the group of apps with privacy-sensitive permissions; that is, apps using such permissions in the Android OS are less successful than apps that do not use such permissions.

⁴⁴ In Chellappa et al. (2010), consumers have heterogeneous privacy concerns, and a monopolist uses technology to reduce the disutility from requesting personal information. The monopolist offers a menu and decides on optimal personalization with and without subsidization. The model’s technology parameter allows accommodating trust-inspiring measures or extending personalization disutility beyond nuisance from targeting to accommodating worries about very sensitive information, or other sources of concern, such as identity theft or embarrassment.

⁴⁵ Similar findings emerge for apps from “top developers” (which receive a visible badge from the Play Store administration) or when we separately analyze corporate apps or apps from large and well-known websites (Facebook, Amtrak, banks, Starbucks, etc.; see Online Appendix Table A7). Note that we do not claim that these coefficients measure the causal relationship. We point out, however, that we control for the mere fact that apps with privacy policies might be published by more experienced suppliers (and possibly are better apps).

⁴⁶ The results could also be explained by differences in the type of content.

⁴⁷ Similar results emerge for games (see Online Appendix Table A7). In this group of apps, users reveal the least sensitive information, and many users are children or young adults.

⁴⁸ Besides a general reluctance to digitize health-related information, users might suspect developers to have incentives for sharing their information with insurance companies, for which health-related information about users and their health-related behaviors is very valuable.

⁴⁹ The insignificant effect for games is in line with additional alternative results, which show that a weaker or even an insignificant relationship between privacy-sensitive permissions and demand exists for games (see Online Appendix Tables A7 and B5).

⁵⁰ That D_{ID} would have a positive coefficient is surprising at first sight, but the permission was hard to recognize for users as intrusive in 2012, because it was called “READ_PHONE_STATE.”

⁵¹ Recall, though, that D_{Ads} has a negative effect when used within variation in our panel data in Table 3.

⁵² For example, a running app needs access to its user's location.

⁵³ We thank one of the anonymous referees for suggesting this additional test.

⁵⁴ Alternatively, we split the sample by apps' past success and verified that the negative relationship between demand and privacy-sensitive permissions is not driven by a specific segment of apps with large or small user bases.

⁵⁵ Note that our data predate increased public awareness of data collection by the U.S. National Security Agency.

⁵⁶ Suggestions include a traffic light scale of an app's intrusiveness (Tsai et al. 2011) or local storage of the user's information on his or her device (Sutanto et al. 2013). Our supporting analysis (see Online Appendix Table B5) indeed indicates that Google's “maliciousness warning” is associated with greater avoidance. However, caution is needed, because such schemes might make it harder for developers to monetize the private information on free apps, which could reduce both the supply and the adoption of apps.

⁵⁷ For example, if many apps introduce permissions at exactly the same stage of their development, and if this were positively correlated with installation or review growth, then we would overestimate the effect.

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